

Partícula libre

$\psi = \eta$ ← onda plana

$$P\eta = \frac{\hbar}{i} \nabla e^{i\left(\bar{k}\cdot\bar{x} - \frac{E}{\hbar}t\right)}$$

$$\uparrow = i \frac{\hbar}{i} \bar{k} e^{i\left(\bar{k}\cdot\bar{x} - \frac{E}{\hbar}t\right)} = \hbar \bar{k} \eta = \bar{p} \eta$$

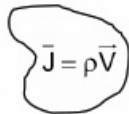
Función
propia

- $\bar{k} = (k_1 \hat{i} + k_2 \hat{j} + k_3 \hat{k})$

- $\bar{x} = (x_1 \hat{i} + x_2 \hat{j} + x_3 \hat{k})$

$$\bar{J}(t, \bar{x}) = \frac{1}{2m} [n^* \bar{p} n + \text{Conj. herm}]$$

$$= \frac{1}{2m} 2\bar{p} |n|^2 = |n|^2 \frac{\bar{p}}{m}$$



$$\bar{J} = \rho \bar{V}$$

(En concordancia con el concepto clásico de densidad de corriente.)

(similar a la ecuación de densidad de corriente eléctrica)

$$\text{Divergencia del gradiente} = - \frac{\hbar^2}{2m} \nabla^2 \psi$$



Ecuación de continuidad

$$\nabla \cdot \vec{J} + \partial \rho = 0$$

$$\rho = \psi^* \psi = |\psi|^2$$

$$J_{(t,\vec{x})} = \frac{1}{2m} \left[\psi^* P \psi + \underbrace{\psi (P \psi)^*}_{\text{(Conj. Hermítico)}} \right]$$

$$J(t, \vec{x}) = \frac{\text{Re}(\psi^* P \psi)}{m}$$

$$P = \frac{\hbar}{i} \nabla$$

Partícula libre

$$\psi = \eta \leftarrow \text{onda plana}$$

$$P\eta = \frac{\hbar}{i} \nabla e^{i\left(\vec{k} \cdot \vec{x} - \frac{E}{\hbar} t\right)}$$

$$\uparrow = i \frac{\hbar}{i} \vec{k} e^{i(\dots)} = \hbar \vec{k} \eta = \vec{p} \eta$$

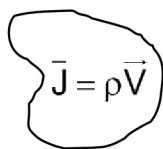
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- $\vec{x} = (x_1 \hat{i} + x_2 \hat{j} + x_3 \hat{k})$

$$\vec{J}(t, \vec{x}) = \frac{1}{2m} [n^* \vec{p} n + \text{Conj. herm}]$$

$$= \frac{1}{2m} 2 \bar{p} |n|^2 = |n|^2 \frac{\bar{p}}{m}$$

ρ



$$\vec{J} = \rho \vec{V}$$

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Teorema de Ehrenfest

$$1) \hat{P}_\psi = m(\dot{\hat{R}}_\psi)$$

$$2) (\dot{\hat{P}}_\psi) = -(\nabla \hat{V})_\psi \quad \nabla \text{ actúa en el operador } V.$$

En 1 dim: $\hat{P}_\psi = m(\dot{\hat{X}}_\psi)$

$$\dot{\hat{P}}_\psi = -\left(\frac{\partial V}{\partial x}\right)_\psi$$

$$(x)_\psi = (\psi, x\psi)$$

$$\frac{d}{dt} x_\psi = \left(\frac{\partial \psi}{\partial t}, x\psi \right) + \left(\psi, \frac{\partial (x\psi)}{\partial t} \right)$$

$$= \left(\frac{\partial \psi}{\partial t}, x\psi \right) + \left(\psi, x \frac{\partial \psi}{\partial t} \right)$$

(x no depende de t) (es el operador posición)

$$= \int_{-\infty}^{\infty} \frac{\partial \psi^*}{\partial t} x \psi dx + \int_{-\infty}^{\infty} \psi^* x \frac{\partial \psi}{\partial t} dx$$

De la ec. de Sch:

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi + \frac{\hbar}{i} \frac{\partial \psi}{\partial t} = 0$$

$$\rightarrow \frac{\partial \psi}{\partial t} = -\frac{i}{\hbar} \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi \right)$$

$$\frac{\partial \psi^*}{\partial t} = \frac{i}{\hbar} \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi^*}{\partial x^2} + V\psi^* \right)$$

$$\rightarrow \frac{d}{dt} x_\psi = \frac{i}{\hbar} \int_{-\infty}^{\infty} \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi^*}{\partial x^2} + V\psi^* \right) x \psi dx +$$

$$-\frac{i}{\hbar} \int_{-\infty}^{\infty} \psi^* x \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi \right) dx$$

$$\frac{d}{dt} x_\psi = \frac{\hbar}{2mi} \int_{-\infty}^{\infty} \frac{\partial^2 \psi^*}{\partial x^2} x \psi dx - \frac{\hbar}{2mi} \int_{-\infty}^{\infty} \psi^* x \frac{\partial^2 \psi}{\partial x^2} dx \quad \dots (\alpha)$$

$(\dot{R}_\psi)$ Derivada temporal del valor medio

$$(\dot{\hat{R}})_\psi = 0 \quad R \text{ no depende del tiempo}$$

$$V = \frac{1}{2} k x^2 \rightarrow \hat{V} = \frac{1}{2} k \hat{X}^2$$

$$\frac{\partial \hat{V}}{\partial x} = \frac{\partial (\frac{1}{2} k x^2)}{\partial x} = kx \rightarrow k\hat{X}$$

$$A = t \times \text{depende del tiempo}$$

$$-\frac{\hbar}{2mi} \int_{-\infty}^{\infty} \psi^* x \frac{a^2 \psi}{ax^2} dx \quad \dots(\alpha)$$

$$\frac{d}{dt} x_{\psi} = \frac{\hbar}{2mi} \int_{-\infty}^{\infty} \frac{\partial^2 \psi^*}{\partial x^2} x \psi dx - \frac{\hbar}{2mi} \int_{-\infty}^{\infty} \psi^* x \frac{a^2 \psi}{ax^2} dx \quad \dots(\alpha)$$

$$\bullet \int_{-\infty}^{\infty} \frac{\partial^2 \psi^*}{\partial x^2} x \psi dx = x \psi \frac{\partial \psi^*}{\partial x} \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \frac{\partial \psi^*}{\partial x} \frac{\partial (x \psi)}{\partial x} dx$$

$$dv = \frac{a^2 \psi^*}{ax^2} dx \rightarrow v = \frac{a \psi^*}{ax}$$

$$\mu = x \psi \rightarrow d\mu = \frac{\partial (x \psi)}{\partial x} dx$$

$$\bullet - \int_{-\infty}^{\infty} \frac{\partial \psi^*}{\partial x} \frac{\partial (x \psi)}{\partial x} dx = - \frac{\partial}{\partial x} (x \psi) \psi^* \Big|_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \psi^* \frac{\partial^2}{\partial x^2} (x \psi) dx$$

$$dv = \frac{\partial \psi^*}{\partial x} dx \rightarrow v = \psi^*$$

$$\mu = \frac{\partial}{\partial x} (x \psi) \rightarrow d\mu = \frac{\partial^2}{\partial x^2} (x \psi) dx$$

$$\bullet \frac{\partial}{\partial x} (x \psi) = \psi + x \frac{\partial \psi}{\partial x}$$

$$\frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} (x \psi) \right) = \frac{\partial \psi}{\partial x} + \frac{\partial \psi}{\partial x} + x \frac{\partial^2 \psi}{\partial x^2}$$

$$\Rightarrow - \int_{-\infty}^{\infty} \frac{\partial \psi^*}{\partial x} \frac{\partial (x \psi)}{\partial x} dx = \int_{-\infty}^{\infty} \psi^* \left(2 \frac{\partial \psi}{\partial x} + x \frac{\partial^2 \psi}{\partial x^2} \right) dx$$

Reemplazando en (α):

$$\frac{d}{dt} x_{\psi} = \frac{\hbar}{2mi} \int_{-\infty}^{\infty} \frac{\partial^2 \psi^*}{\partial x^2} x \psi dx - \frac{\hbar}{2mi} \int_{-\infty}^{\infty} \psi^* x \frac{a^2 \psi}{ax^2} dx \quad \dots(\alpha)$$

$$\left(\dot{x}_{\psi} \right) = \frac{\hbar}{2mi} \int_{-\infty}^{\infty} \psi^* 2 \frac{\partial \psi}{\partial x} dx = \frac{1}{m} \int_{-\infty}^{\infty} \psi^* P_{\psi} dx$$

$$= \frac{1}{m} P_{\psi}$$

$$\dot{x}_{\psi} = \frac{P_{\psi}}{m}$$

$$x_{\psi} = \int_{-\infty}^{\infty} \psi^* x \psi dx$$

Obs: $\left(\frac{d}{dt}\right)_\psi = 0$ (x no depende de t)

$$\dot{x}_\psi = \frac{d}{dt} \int_{-\infty}^{\infty} \psi^* x \psi dx$$

Commutador: $[\hat{f}, \hat{g}] = \hat{f}\hat{g} - \hat{g}\hat{f}$
 $[\hat{x}, \hat{p}] = ?$

$$[\hat{x}, \hat{p}]\psi = \hat{x}\hat{p}\psi - \hat{p}(\hat{x}\psi)$$

$$= x \frac{\hbar}{i} \frac{d\psi}{dx} - \frac{\hbar}{i} \frac{d}{dx}(x\psi)$$

$$= x \cancel{\frac{\hbar}{i} \frac{d\psi}{dx}} - \frac{\hbar}{i} \left(\psi + x \cancel{\frac{d\psi}{dx}} \right)$$

$$[\hat{x}, \hat{p}]\psi = -\frac{\hbar}{i} \psi \rightarrow \boxed{[\hat{x}, \hat{p}] = i\hbar}$$

No conmutan

No son medibles

Simultáneamente

Dos operadores conmutan
cuando $[\hat{f}, \hat{g}] = 0$

($\exists$ una base común)